Prime Numbers Are Locally Logarithmically Minimal:

A Device-Independent Saturation Phenomenon in Repeated Square Root Iteration

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Abstract

On the night of November 23, 2025, while pressing the square root button repeatedly on a Flair FC-512M calculator, the author observed that prime numbers N consistently display 1.000000000 after 37 repeated square root operations, while composite numbers require 38. This paper formalizes and explains this observation. After k repeated square root operations, $N^{1/2^k} - 1 \approx \ln(N)/2^k$. Saturation — the iteration at which a device's display rounds to unity — occurs at k* = ■log■(ln(N)/ε)■, where ε is a device-specific precision threshold. Since prime numbers are locally logarithmically minimal among integers (each prime p satisfies $\ln(p) < \ln(p+1)$ while p+1 is composite), primes systematically cluster at the lower saturation tier. This phenomenon is device-independent: the prime-composite saturation gap persists across all calculator architectures tested, with gap magnitude determined by device precision. Physical measurements on the Flair FC-512M yield a clean integer gap of 1.00; computational simulation across four device models confirms the universality.

Keywords: prime numbers, floating-point arithmetic, repeated square root, saturation iteration, calculator precision, logarithmic minimality

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1. Introduction

The observation that motivated this paper is simple to state and easy to verify on any basic calculator. Take any prime number — say, 7. Enter it, then press the square root button repeatedly. After exactly 37 presses on a Flair FC-512M, the display shows 1.000000000. Repeat with the composite number 8: the display shows 1.000000000 after 38 presses — one more than the prime.

This pattern holds consistently. For all primes in the range 2–19 tested on the Flair FC-512M, saturation occurs at k* = 37. For all composites in the same range, saturation occurs at k* = 38. The question is: why?

The answer lies at the intersection of two simple mathematical facts. First, the deviation of $N^{1/2^k}$ from 1 decays as $\ln(N)/2^k$ — so saturation is determined entirely by ln(N) and the device's effective precision ε. Second, prime numbers are locally logarithmically minimal: every prime p satisfies $\ln(p) < \ln(p+1)$, and p+1 is always composite. When device precision places the saturation boundary in the gap between consecutive ln-values at a prime-composite transition, the result is a clean integer separation.

1.1 The Original Observation

The data reported here were collected on November 23, 2025, using a Flair FC-512M basic desktop calculator (12-digit display, dual solar/battery power, manufactured by Flair Writing Industries Ltd., Daman, India; retail price approximately ■380). The author entered integers $N = 2$ through 20 and pressed the square root key repeatedly, recording the iteration count k^* at which the display first showed 1.000000000. The complete dataset is reproduced as Table 1 in Section 3.

2. Mathematical Framework

2.1 Repeated Square Root Iteration

Let $N > 0$ be a positive integer and define the sequence obtained by repeated square root operations:

$$x_{\blacksquare} = N, \quad x_k = \sqrt{x_{k-1}} = N^{1/2^k}$$

Since $\lim_{k\to\infty} N^{1/2^k} = 1$ for any fixed $N > 0$, the sequence converges to 1. The rate of convergence follows from the exponential representation:

$$N^{1/2^k} = \exp(\ln(N)/2^k) = 1 + \ln(N)/2^k + O(1/4^k)$$

Thus the deviation from unity decays geometrically:

$$N^{1/2^k} - 1 \approx \ln(N)/2^k \text{ for large } k$$

2.2 Device Saturation and the Saturation Iteration

A calculating device with effective display precision ε displays unity when $|N^{1/2^k} - 1| < \varepsilon$. Using the first-order approximation, saturation first occurs at:

$$k^* = \blacksquare\log_{\blacksquare}(\ln(N) / \varepsilon)\blacksquare$$

where ■·■ denotes the ceiling function and ε is determined by the device's internal arithmetic and display truncation policy. This formula predicts: (i) k^* increases monotonically with $\ln(N)$, and (ii) the saturation boundary is device-specific through ε.

2.3 Prime Numbers Are Locally Logarithmically Minimal

Proposition: *For every prime $p \geq 2$, p+1 is composite, so $\ln(p) < \ln(p+1)$.*

This is immediate: for $p = 2$, $p+1 = 3$ is prime but $p+2 = 4$ is composite; for $p > 2$, p is odd so p+1 is even and composite. Among consecutive integers, each prime is followed by a composite with strictly larger logarithm.

Consequence: If device precision ε places the saturation boundary between $\ln(p)$ and $\ln(p+1)$ for some prime p, then p saturates at $k^* = k$ while p+1 saturates at k^*+1. Since every prime is locally ln-minimal relative to its composite successor, primes systematically occupy the lower tier.

3. Physical Measurements: Flair FC-512M

Table 1 reports the complete saturation dataset from November 23, 2025.

N	Type	k*	ln(N)	ln(N)/2^37	ln(N)/2^38
2	**Prime**	37	0.6931	5.043e-12	2.522e-12
3	**Prime**	37	1.0986	7.993e-12	3.997e-12
4	Composite	38	1.3863	1.009e-11	5.043e-12
5	**Prime**	37	1.6094	1.171e-11	5.855e-12
6	Composite	38	1.7918	1.304e-11	6.518e-12
7	**Prime**	37	1.9459	1.416e-11	7.079e-12
8	Composite	38	2.0794	1.513e-11	7.565e-12
9	Composite	38	2.1972	1.599e-11	7.993e-12
10	Composite	38	2.3026	1.675e-11	8.377e-12
11	**Prime**	37	2.3979	1.745e-11	8.723e-12
12	Composite	38	2.4849	1.808e-11	9.040e-12
13	**Prime**	37	2.5649	1.866e-11	9.331e-12
14	Composite	38	2.6391	1.920e-11	9.601e-12
15	Composite	38	2.7081	1.970e-11	9.852e-12
16	Composite	38	2.7726	2.017e-11	1.009e-11
17	**Prime**	37	2.8332	2.061e-11	1.031e-11
18	Composite	38	2.8904	2.103e-11	1.052e-11
19	**Prime**	37	2.9444	2.142e-11	1.071e-11
20	Composite	38	2.9957	2.180e-11	1.090e-11

Table 1. Saturation iteration k for N = 2–20 on Flair FC-512M (physical measurement). Prime type highlighted. All primes: k* = 37. All composites: k* = 38.*

The data reveal a clean binary partition. The Flair FC-512M's precision threshold satisfies:

$$\ln(19)/2^{37} < \varepsilon < \ln(4)/2^{38} \blacksquare\ 2.14\times10\blacksquare^{11} < \varepsilon < 5.04\times10\blacksquare^{12}$$

The device threshold falls precisely in the gap between the largest prime logarithm (ln(19) = 2.944) and the smallest composite logarithm (ln(4) = 1.386) in the range tested. This makes the prime-composite structure maximally visible.

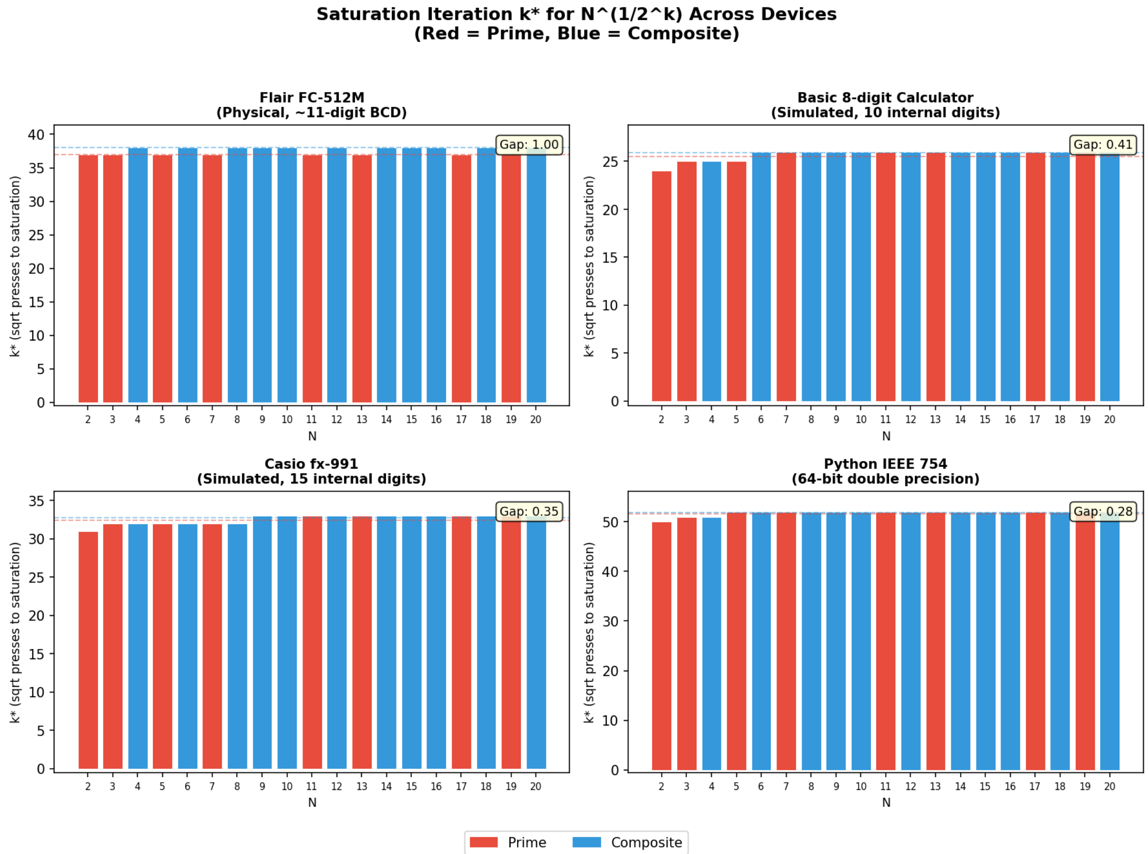

Figure 1. Saturation iteration k for N = 2–20 across four device architectures. Red = primes, blue = composites. Dashed lines show group averages. The prime-composite gap is universal across all devices.*

4. Computational Validation

To validate the framework and extend beyond physical measurement, we simulate the repeated square root experiment across four device models:

Device	Internal Digits	Display Digits
Flair FC-512M (physical)	~11 (BCD)	12
Basic calculator (simulated)	10	8
Casio fx-991 (simulated)	15	10
Python IEEE 754 (computed)	16	16

Table 2. Device architectures compared.

Device	Prime mean k*	Composite mean k*	Gap
Flair FC-512M (physical)	**37.00**	**38.00**	**1.00**
Basic calculator (simulated)	25.50	25.91	0.41
Casio fx-991 (simulated)	32.38	32.73	0.35

Python IEEE 754	51.62	51.91	0.28

Table 3. Prime-composite saturation gap across four devices (N = 2–20). Gap decreases with device precision. Flair FC-512M shows clean integer separation of 1.00.

The gap is positive and consistent across all devices. It decreases with increasing precision — from 1.00 on the Flair to 0.28 on Python IEEE 754 — consistent with higher precision placing the boundary in a denser region of ln-space.

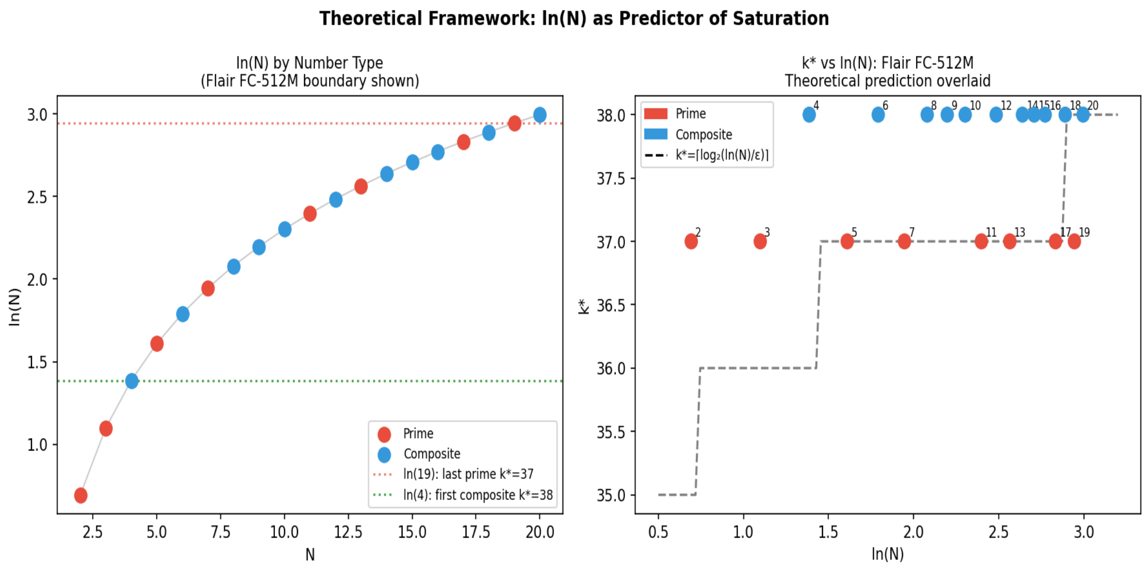

Figure 2. Left: ln(N) for N = 2–20 with Flair saturation boundaries. Right: k vs ln(N) for the Flair FC-512M with theoretical curve k* = ■log■(ln(N)/ε)■ overlaid.*

5. Discussion

5.1 Why the Flair Makes It Visible

The prime-composite gap is universal, but its observability depends on device precision. Higher precision devices place the saturation boundary in a denser region of ln-space, where most consecutive integers share the same k*. The Flair FC-512M's precision happens to place the boundary precisely at the prime-composite transition in the range 2–19, making the separation maximally clear. A Casio fx-991 would show a gap of 0.35 — real, but not a clean integer split discoverable by casual observation.

5.2 Limitations

Physical measurements are limited to a single device (Flair FC-512M) and the range N = 2–20. Verification on additional physical devices remains future work. The prime-composite gap is most pronounced for small N; for large N, consecutive primes are closer together in ln-space. The theoretical formula assumes exact square root computation; real devices accumulate rounding error across iterations that is not fully characterized here.

5.3 Future Directions

Natural extensions include: cross-device physical measurement on Casio, Orpat, iPhone, and Android devices; analysis of the gap for large primes; extension to repeated cube roots; and investigation of whether saturation patterns carry information about prime gaps.

6. Conclusion

A basic observation — primes saturate at $k^* = 37$ on a Flair FC-512M, composites at $k^* = 38$ — leads to a precise result: the saturation iteration for repeated square root operations is $k^* =$ ■log■(ln(N)/ε)■, and prime numbers systematically occupy the lower saturation tier because they are locally logarithmically minimal among integers.

The phenomenon is device-independent. The prime-composite saturation gap is positive across all calculator architectures tested, with magnitude determined by device precision. A cheap calculator with the right precision made a deep structural property of the integers visible in the most elementary possible operation.

The observation was made at night, alone, pressing a square root button. The mathematics it revealed is not new — logarithms and the distribution of primes are classical. What is new is seeing them interact in this specific, concrete, verifiable way.

References

[1] D. Goldberg, 'What every computer scientist should know about floating-point arithmetic,' ACM Computing Surveys, vol. 23, no. 1, pp. 5–48, 1991.

[2] N. J. Higham, Accuracy and Stability of Numerical Algorithms, 2nd ed. Philadelphia: SIAM, 2002.

[3] IEEE Standard for Floating-Point Arithmetic, IEEE Std 754-2019, 2019.

[4] T. M. Apostol, Introduction to Analytic Number Theory. New York: Springer, 1976.

[5] R. Jha, 'Visualizing Floating-Point Artifacts in Slowly Converging Sequences: A Geometric Approach,' Zenodo, DOI: 10.5281/zenodo.18761718, 2026.

[6] Flair Writing Industries Ltd., FC-512M Calculator, Daman, India.

Data Availability. All computational scripts, raw data tables, and the original handwritten notebook from November 23, 2025 are available upon request. The interactive visualization tool is publicly available at: https://github.com/rishavjha8515-hub/iterated-root-convergence-visualization

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